EUR Research Information Portal

Empowering leadership and job crafting: The role of employee optimism

Published in:
Stress and Health

Publication status and date:
Published: 01/01/2018

DOI (link to publisher):
10.1002/smi.2818

Document Version
Publisher's PDF, also known as Version of record

Document License/Available under:
Article 25fa Dutch Copyright Act

Citation for the published version (APA):
Thun, S., & Bakker, A. (2018). Empowering leadership and job crafting: The role of employee optimism. *Stress and Health*, *34*, 573-581. https://doi.org/10.1002/smi.2818

Link to publication on the EUR Research Information Portal

Terms and Conditions of Use
Except as permitted by the applicable copyright law, you may not reproduce or make this material available to any third party without the prior written permission from the copyright holder(s). Copyright law allows the following uses of this material without prior permission:

- you may download, save and print a copy of this material for your personal use only;
- you may share the EUR portal link to this material.

In case the material is published with an open access license (e.g. a Creative Commons (CC) license), other uses may be allowed. Please check the terms and conditions of the specific license.

Take-down policy
If you believe that this material infringes your copyright and/or any other intellectual property rights, you may request its removal by contacting us at the following email address: openaccess.library@eur.nl. Please provide us with all the relevant information, including the reasons why you believe any of your rights have been infringed. In case of a legitimate complaint, we will make the material inaccessible and/or remove it from the website.

Erasmus University Rotterdam
Making Minds Matter



Received: 17 July 2017 | Revised: 8 March 2018 | Accepted: 23 April 2018

DOI: 10.1002/smi.2818

RESEARCH ARTICLE

WILEY

Empowering leadership and job crafting: The role of employee optimism

Sylvi Thun[1,2] | Arnold B. Bakker[3,4]

[1] Trøndelag R&D Institute, Trondheim, Norway

[2] Department of Research and Development, Division of Psychiatry, St. Olav's University Hospital, Trondheim, Norway

[3] Center of Excellence for Positive Organizational Psychology, Erasmus University Rotterdam, Rotterdam, The Netherlands

[4] University of Johannesburg, Johannesburg, South Africa

Correspondence
Sylvi Thun, Trøndelag R&D Institute, Trondheim, Norway.
Email: sylvi.thun@tfou.no

Abstract

The objective of this study was to test the relationship between empowering leadership and job crafting and to examine the moderating role of optimism as a personal resource. We hypothesized that the association between empowering leadership and job crafting would be stronger for employees with high (vs. low) levels of optimism. A total of 331 Norwegian workers from a variety of occupations participated in our study. Results of structural equation modelling analysis generally supported our hypotheses. Empowering leadership was positively related to 3 of the 4 job crafting strategies investigated (increasing structural job resources, increasing social job resources, and increasing challenging job demands; but not reducing hindrance job demands). Moreover, as hypothesized, optimism strengthened the empowering leadership–job crafting relationship for increasing structural resources and increasing challenging demands. The results suggest that empowering leadership is an important antecedent of job crafting strategies, except for reducing hindrance demands. The implications of these findings are discussed.

KEYWORDS

empowering leadership, JD–R theory, job crafting, optimism, personal resources

1 | INTRODUCTION

To remain successful and attractive in the ever-changing world of work, it is crucial for organizations to attract and employ individuals who are able to deal with the ongoing changes and challenges. Changes in the work structure have led to an increased focus on organization and employee capabilities for making adjustments (van den Heuvel, Demerouti, & Peeters, 2015). One way to handle changes and challenges is job crafting (JC), the process where employees act as active agents, shaping, redefining, and creating their job to ensure a good person–job fit in their work environment (Tims, Bakker, & Derks, 2012; Grant & Parker, 2009; Tims, Bakker, & Derks, 2014; Wrzesniewski & Dutton, 2001). Another important variable affecting the person–job fit and the work environment is leadership. As leaders play a crucial role in the social context of work, organizations will profit if leaders inspire their employees to use strategies encouraging development such as JC, which fosters engagement and good work attendance (Petrou, Demerouti, Peeters, Schaufeli, & Hetland, 2012). Research on JC has increased rapidly during the past decade. However, the literature has not thoroughly explored how leader behaviours relate to JC (Rudolph, Katz, Lavigne, & Zacher, 2017; Wang, Demerouti, & LeBlanc, 2017) and how leader behaviours interact with individual differences (Schaufeli & Taris, 2014; Spreitzer, 1995). Therefore, the aim of this study is to investigate the relation between JC and leadership, and the moderating impact of optimism, the global expectation that good things will happen in the future (Scheier & Carver, 1985). As human behavior often results from an interaction between personal and environmental factors (Schaufeli & Taris, 2014), we want to contribute new knowledge to extend the theoretical field of JC.

There are several important reasons for investigating JC. JC improves the fit between the person and the job and is positively related to work engagement, job satisfaction, meaningfulness, and job performance (Demerouti, 2014; Rudolph et al., 2017). Evidence from recent intervention studies shows that job crafters increase their affective well-being and job performance (Gordon et al., 2018; van

 Copyright © 2018 John Wiley & Sons, Ltd.

den Heuvel et al., 2015; van Wingerden, Bakker, & Derks, 2017; van Wingerden, Derks, & Bakker, 2017). When employees craft their jobs, they make changes in tasks or relationships, build new job resources, and reduce their hindering demands (van den Heuvel et al., 2015). Job crafters can make changes in their resources or demands to fit their personal needs and abilities better; employees can seek challenges (e.g., starting a new project), increase resources (e.g., asking for more feedback), and reduce their demands (e.g., minimizing demanding aspects of the job) (Tims, Bakker, & Derks, 2012). If people optimize their work environment, they create a fit between person and job, which has a positive impact on well-being, meaningfulness, and performance (e.g., Bakker, 2017; Tims, Derks, & Bakker, 2016). Because previous research has shown that JC has positive effects, we need to study how we can foster JC.

Leadership may provide employees with the resources, legitimate reasons, or freedom to employ JC (Wang, Demerouti, & Bakker, 2017), and several leadership styles may be good candidates as antecedents of JC. To the best of our knowledge, only three studies have explored leadership in relation to JC. Slemp, Kern, and Vella-Brodrick (2015) found that leader autonomy support is positively related to JC. Wang and his colleagues (2017) reported a positive relationship between transformational leadership and JC. Esteves and Lopes (2017) investigated Pearce, Sims, and Kivlighan's (2002) leadership model based on five behavioural leadership styles in relation to JC. They found that empowering leadership was strongly related with the increase of challenging demands and the increase of social resources. Together, these studies highlight that leaders' behaviour has implications for employees' JC. Accordingly, a question arises as to *how* employees react to work environments where self-initiated changes are encouraged.

Although the abovementioned leadership styles are relevant, we argue that empowering leadership has a particularly promising potential, because it specifically aims at making people feel more confident to take charge of their own work activities (Ahearne, Mathieu, & Rapp, 2005). A focus on empowering leadership can give new insight into the determinants of JC as leaders are in a good position to boost JC by acting as a role model or by encouraging employees to craft their job (Wang, Demerouti, & Bakker, 2017). Furthermore, this approach would contribute knowledge important for the theoretical development of the job demands–resources (JD–R) perspective of JC. In the JD–R conceptualization of JC, more studies are needed to understand what influences JC (Bakker et al., 2012; Bipp & Demerouti, 2015; Rudolph et al., 2017). According to Wang, Demerouti, and Bakker (2017), the perspective on leadership and employee JC has received much less research attention than the focus on individual difference factors (e.g., Bakker et al., 2012; Bipp & Demerouti, 2015) or job characteristics (e.g., Petrou et al., 2012). In addition, we want to follow-up on Spreitzer's (1995) request to examine the behavioural consequences of empowerment and the moderators (individual differences) of the link between empowerment and organizational behaviour.

To develop JC theory further, more research on the role of moderators is needed (Rudolph et al., 2017; Tims & Bakker, 2010), especially in order to investigate whether personal resources can act as a buffer in the motivational path in the JD–R conceptualization of JC. By acknowledging JC dynamics, we have a unique opportunity to maximize employee potential and achieve positive organizational outcomes (Petrou et al., 2012). By gaining more knowledge about JC, its antecedents and moderators, leaders may also be able to prevent the possible negative consequences of JC (Tims et al., 2016). In our study, we also answer the call by Rudolph et al. (2017), arguing consideration of characteristics that might foster or undermine employee JC. By focusing on possible moderators, we gain knowledge on when it is possible to increase the use of JC and on the development of a JC theory. Our hypothesized model builds on the theoretical framework of JD–R in combination with the principles of empowering leadership and optimism.

2 | JC THROUGH THE LENS OF JD–R THEORY

Historically, the content of JC was illustrated by Kulik, Oldham, and Hackman (1987) when they suggested that individuals may redesign their jobs on their own initiative. They argued that work redesign is a participative change process. The term "job crafting" was coined by Wrzesniewski and Dutton (2001), who defined JC as "the physical and cognitive changes individuals make in their task or relational boundaries at work" (p. 179). According to the JD–R approach, JC is conceptualized as proactive behaviours through which employees change their levels of job demands and job resources, that is the characteristics of their work to gain a better person–job fit (Petrou et al., 2012; Tims et al., 2012; Tims et al., 2014). Consequently, crafting job resources could take the form of increasing social resources meaning development of employees' relations with colleagues/nearest leader including support, and/or coaching/feedback from leaders or colleagues. Another way to craft job resources is by increasing structural resources by developing competence and professional skills in the work context. Crafting job demands could take the form of increasing challenging demands which consist of workers' active search for new challenging work tasks (Tims et al., 2012). The last dimension of JC refers to reducing hindering demands, for instance, avoiding mentally demanding tasks. When using the latter dimension, employees can adjust the job by simplifying it, if it becomes too demanding (Clegg & Spencer, 2007).

JD–R theory outlines how job characteristics influence occupational well-being and job performance (Bakker & Demerouti, 2007; Bakker & Demerouti, 2014; Bakker & Demerouti, 2017; Demerouti, Bakker, Nachreiner, & Schaufeli, 2001). Accordingly, all job characteristics are part of one of two main categories: job demands and job resources. Whereas job demands can be stressful and may have an indirect negative impact on performance, job resources help to deal with job demands and motivate employees, indirectly facilitating job performance. Another important mechanism in the JD–R theory is personal resources as it influences the perception and availability of job resources (Xanthopoulou, Bakker, Demerouti, & Schaufeli, 2009). Personal resources refer to positive self-evaluations that are linked to resiliency and the individual's sense of their ability to control and impact upon their environment successfully (Hobfoll, Johnson, Ennis, & Jackson, 2003). Furthermore, JC is argued to be an important mechanism in JD–R theory as it connects job characteristics to work

outcomes (Tims & Bakker, 2010). Although JC is a proactive bottom-up process, it is crucial that job crafters perceive and experience necessary work conditions to make JC effective. One way to inspire more use of JC is through leadership (Petrou et al., 2012).

3 | EMPOWERING LEADERSHIP AND JC

Empowering leadership has been found to increase employee proactive behaviour and may therefore be positively related to JC (Esteves & Lopes, 2017; Wang, Demerouti, & Le Blanc, 2017). Empowering leadership is defined as a process where the leader transfers power from himself/herself to employees and gives them strength to make their own decisions by providing additional responsibility, decision-making authority over work, and resources (Ahearne et al., 2005). Empowering leaders aim to give employees a feeling of meaning in their work, giving them time and latitude to decide for themselves how to perform tasks and the support needed to handle additional responsibility effectively, encouraging self-development (Ahearne et al., 2005; Zhang & Bartol, 2010). They inspire employees to lead themselves and motivate development of self-leadership skills to contribute more fully to the organization (Sims, Faraj, & Yun, 2009; Zhang & Bartol, 2010).

Self-determination theory (Gagné & Deci, 2005) proposes that the needs for autonomy and competence are crucial to the formation of intrinsic motivation, which encourages creative, proactive, and self-directed activities. In addition, self-determination theory proposes that managers' interpersonal style is a factor in the social context that facilitates intrinsic motivation. Martin, Liao, and Campbell (2013) argue that the characteristics of empowering (leadership) behaviours facilitate employees in gaining a sense of autonomy and competence. A higher sense of autonomy and competence enhances intrinsic motivation and can result in more proactive behaviour, including JC. Leaders may affect the process of employee JC as empowering leadership encourages employees to spend more time exploring how to perform their core tasks which can subsequently make JC beneficial for both the individual and the organization (Martin et al., 2013; Wang, Demerouti, & Le Blanc, 2017).

Lyons (2008) argues that the decision to craft the job is influenced by the perceived opportunity to do so. Leaders have a unique possibility of empowering followers to use JC and can enlighten how followers can make their own jobs more satisfying, meaningful, and engaging (Bakker & Demerouti, 2014; Bakker & Demerouti, 2017). The perception of an empowering leader relates positively to an increase in challenges in the work environment and an increase in social resources (Esteves & Lopes, 2017). Petrou et al. (2012) argue that empowering leadership may encourage and facilitate JC by creating an "active learning" climate that fosters personal initiative. If the leader empowers their employees, these employees can feel they are entitled and able to enact their autonomy and start to engage in strategies encouraging a good person–job fit. The more reassurance and assistance in building competence, autonomy, and flexibility that employees perceive, the more likely they are to use JC (Ahearne et al., 2005; Clegg & Spencer, 2007; Demerouti, 2014; Wrzesniewski & Dutton, 2001). Because empowering leaders encourage proactive behaviour (Ahearne et al., 2005; Esteves & Lopes, 2017; Sims et al., 2009; Wang, Demerouti, & Le Blanc, 2017; Zhang & Bartol, 2010), we predict:

> **Hypothesis 1.** *Empowering leadership is positively related to job crafting in the form of (a) increasing social and structural job resources; (b) increasing job challenges and decreasing hindering job demands.*

4 | OPTIMISM AND JC

Previous research has documented that optimism is relevant for people's psychosocial adaptation and well-being (Alarcon, Bowling, & Khazon, 2013; Krok, 2015) and related to positive outcomes across a broad range of situations (Carver, Scheier, & Segerstrom, 2010; Rasmussen, Scheier, & Greenhouse, 2009). According to Scheier and Carver (1985), these positive global expectancies, referred to as dispositional optimism, are relatively stable across time and context and constitute the opposite of dispositional pessimism. Individuals, who see desired outcomes as achievable, continue to strive towards those outcomes, even when progress becomes challenging (Scheier & Carver, 1992). Models of proactivity suggest that personal characteristics and beliefs serve as individual difference and are therefore a possible antecedent of JC (Grant & Ashford, 2008; Grant & Parker, 2009). According to Hobfoll (1989), personal resources such as optimism could affect how employees adapt to work changes. Recently, two theoretical advancements of JC inside the JD–R perspective illustrate that a variety of personal resources are associated with JC (Demerouti, 2014; Wang, Demerouti, & Le Blanc, 2017). However, Schaufeli and Taris (2014) argue that the exact role personal resources play in the JD–R theory is not perfectly clear and needs more empirical investigation.

Until now, we know that personal resources can help individuals to deal with challenges and can boost the impact of challenging job demands (Bakker & Sanz-Vergel, 2013). Optimism also predicts opportunities for self-development and later work engagement (Xanthopoulou et al., 2009). Optimistic individuals invest effort and commitment in goal pursuit (Monzani et al., 2015). As optimism goes along with self-development and one's range of control, the optimistic employee might be more open to leaders' encouraging behaviours. Optimism can offer employees the confidence needed to take initiatives and to change their work environment. Hence,

> **Hypothesis 2.** *Optimism is positively related to job crafting in the form of (a) increasing social and structural job resources; (b) increasing job challenges and decreasing hindering demands.*

5 | THE MODERATING ROLE OF OPTIMISM

Optimistic individuals are known to take a proactive approach to health promoting activities and to employ effort towards favourable outcomes (Carver & Scheier, 2014). Compared with pessimistic individuals, optimistic individuals are more likely to invest, act, and put

effort into achieving their own goals. Specifically, optimistic people tend to engage in practices that are more positive and are also more likely to use active coping strategies (Brydon, Walker, Wawrzyniak, Chart, & Steptoe, 2009; Scheier & Carver, 1992). Furthermore, Xanthopoulou et al. (2009) conceptualize that optimism affects employees' functioning at work. The degree of optimism may also determine the way people comprehend the environment, formulate it, and react to it (Judge, Locke, & Durham, 1997; Xanthopoulou et al., 2009). High optimism predicts high effort and success as optimistic persons perceive challenges as an opportunity for personal growth. People's perception of and adaptation to environments depend on their levels of personal resources (Bandura, 2000). In other words, personal resources such as optimism may function as moderators (Schaufeli & Taris, 2014; Wang, Demerouti, & Le Blanc, 2017). According to JD–R theory, personal resources may—just like job resources—facilitate the motivational impact of, for instance, challenging job demands (Bakker & Sanz-Vergel, 2013) and optimism is functional in achieving work goals (Bakker & Demerouti, 2014; Bakker & Demerouti, 2017; Tims & Bakker, 2010). Optimistic employees will be most likely to effectively use the additional responsibility, decision-making authority over work, and resources granted to them by empowering leaders. Combining the core principles of empowering leadership and integrating it with the principles of optimism and JD–R theory, we propose the final hypothesis:

> **Hypothesis 3.** *Optimism moderates the relationship between empowering leadership and job crafting in the form of (a) increasing social and structural job resources; and (b) increasing job challenges and decreasing hindering demands. These relationships will be stronger for followers high versus low in optimism.*

6 | METHOD

6.1 | Procedure

A cross-sectional study was carried out among Norwegian employees (N = 331) in March 2015. Participants received an e-mail with a link to a web-based questionnaire. They were asked to provide their responses anonymously. Inclusion criteria were that the participants' main daily activity was work, and that they had to be at least 18 years old. All survey items were formulated in Norwegian. A signed informed consent form was required before participation. The Norwegian Social Science Data Service approved the study.

The first author of this paper was the project manager. Thirty-six assistants were responsible for the data collection and had to recruit 10 respondents each, five men and five women. Steps were taken to ensure age variance in the sample. Each assistant had to recruit two of 10 respondents in the age range 18–29 years; four of 10 in the age range 30–44; three of 10 in the age range 45–59 years; and one of 10 in the age range 60–74 years. Additionally, three or four of the 10 respondents had to work in the public sector and six or seven in the private sector; two of 10 had to work part-time, and seven had to work full-time. These specific criteria were based on the labour force survey provided by Statistics Norway (2015).

6.2 | Participants

The sample in this study consisted of 166 females and 164 males (one missing). Participants' ages ranged from 20 to 69 years, the average age was 41.31 years (SD = 13.22). From the 360 employees asked to participate, 331 completed the survey (response rate is 92%). Most participants (89%) had permanent work, and 58% worked in the private sector. The participants came from a variety of business sectors (n = 244, 26.3% did not respond to this question). Among the participants, there were, for instance, health care workers (19.3%), lecturers and researchers (13.9%), and there were participants from mining and energy production (1.5%), manufacturing (5.1%), transport and communication (1.5%), financial and insurance services (5.1%), public administration (6.9%), building and construction (5.7%), commodity trade (10.9%), hotel and restaurant (3%), and primary industries (0.6%).

6.3 | Measures

6.3.1 | Job crafting

The JC instrument used in this survey was the original 21-item JC scale (Tims et al., 2012). The items were translated to Norwegian and then back translated to English to ensure adequate translation by following Brislin's (1970) recommendations. Accordingly, we measured four JC strategies: (a) increasing structural job resources (five items, e.g., "I try to develop my capabilities"), (b) increasing social job resources (five items, e.g., "I ask my supervisor to coach me"), (c) increasing challenging job demands (five items, e.g., "When an interesting project comes along, I offer myself proactively as project coworker"), and (d) hindrance job demands (six items, e.g., "I make sure that my work is less mentally intense"). Answers were given on a 5-point scale ranging from (1) *never* to (5) *very often*. Cronbach's α for the increasing structural job resources was .80, for increasing social job resources .75, for increasing challenging demands .79, and for hindrance job demands .80. Cronbach's α for a total JC measure was .84.

6.3.2 | Empowering leadership

The instrument of empowering leadership was retrieved from the General Nordic Questionnaire for Psychological and Social Factors at Work (Wannström, Peterson, Asberg, Nygren, & Gustavsson, 2009). The measure contains questions regarding encouraging employees to participate in decision-making and to express their differing opinions and helping individual development at work (three items, e.g., "Does your supervisor encourage you to participate in taking important decisions?"). Participants responded on a 5-point scale ranging from (1) *hardly ever/never* to (5) *often/always*. Cronbach's α for the scale was .86.

6.3.3 | Optimism

We measured optimism with the Life Orientation Test—Revised (Scheier, Carver, & Bridges, 1994). The instrument consists of items measuring generalized optimism versus pessimism. As this study was not interested in the nature of optimism versus pessimism, only the items measuring optimism were included (three items). An example item is, "In uncertain times, I usually expect the best" (1 = *I disagree*, 5 = *I agree*). Cronbach's α was .80.

6.4 | Statistical analysis

To test our hypotheses, we conducted structural equation modelling (SEM) analyses using IBM SPSS AMOS version 25. We preferred to run the analysis including the moderation analysis in SEM rather than using hierarchical regression analyses as SEM allows for assessing and correcting for measurement error. Additionally, it provides measures of model fit (Byrne, 2016). The interaction term Empowering Leadership × Optimism was computed by centring empowering leadership and optimism, followed by multiplying the variables. To counteract multicollinearity, we created a centred product term in IBM SPSS version 24 before entering the data in the AMOS software. We tested one model that included empowering leadership, optimism, the interaction term Empowering Leadership × Optimism, and the four JC strategies. The hypothesized main effects and interaction effect were simultaneously included in the model testing. The model fit was assessed with a selection of fit values, including Bentler comparative fit index (CFI), Tuckler Lewis index (TLI), goodness of fit index (GFI), and root-mean-square error of approximation (RMSEA). According to the guidelines by Hu and Bentler (1999), goodness of fit values exceeding 0.95 and RMSEA values of 0.06 or less indicate a good model fit.

Prior to the analyses, the dataset was screened for missing data. The number of missing data did not exceed 8%. Before testing the hypotheses, we conducted confirmatory factor analysis for our proposed four factor structure and three alternative models in IBM SPSS AMOS. We inspected the modification indices of the model and adjusted the model in terms of the JC theory (Tims et al., 2012). Because AMOS does not provide modification indices when data are missing, missing data were replaced using maximum likelihood estimation.

7 | RESULTS

7.1 | Descriptive results

Table 1 presents the means, standard deviations, correlations, and internal consistencies of all scales included in this study. The participants reported highest mean score for JC structural resources (M = 4.40, SD = 0.60) and the lowest mean score for JC hindrance demands (M = 2.32, SD = 0.73). The internal consistencies of the scales are generally good as Cronbach's αs are all above .75. Prior to the hypotheses testing, tests of the measurement model and alternative models were conducted to control for common method variance and to establish discriminative validity (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003). The one-, two-, and three-factor models exhibited poorer fit to the data than the proposed four-factor model (Table 2). The testing of alternative models revealed that the four-factor structure of the research variables was consistent with the conceptual model and that the manifest variables loaded on the intended latent variables. The results of the CFA corresponded with the four-factor structure reported in the original validity study (Tims et al., 2012). Fit indices of the four-factor model conducted through a CFA were χ^2 = 367.76, df = 178, p < .001, χ^2/d = 2.07, CFI = 0.93, TLI = 0.91, GFI = 0.90, RMSEA = 0.06. The model had a satisfactory fit (Hu & Bentler, 1999). All items loaded significantly on the latent variables, with coefficients ranging from .40 to .93 with one exception (the last item of increasing structural resources) that had a factor loading of .17 (p = .003).

7.2 | Hypotheses testing

Hypotheses 1a and 1b state that empowering leadership is positively related to JC, and Hypotheses 2a and 2b propose that optimism is positively related to JC. Table 3 presents the results of the SEM analysis. The hypothesized model showed a good to excellent fit to the data: χ^2 = 3.44, df = 2, χ^2/df = 1.72, TLI = 0.935, CFI = 0.995, RMSEA = 0.047 (Hu & Bentler, 1999). Hypotheses 1a and 2a were supported as empowering leadership and optimism had a significant positive relation to JC structural resources and JC social resources. Hypotheses 1b and 2b were partly supported as empowering leadership and optimism had a significant positive relation to JC challenging demands, and optimism had a significant relation to JC hindrance demands, however in the opposite direction than was hypothesized.

Furthermore, results of the SEM analysis indicated that empowering leadership was a stronger predictor of JC structural resources (β = .36, p < .001) and JC challenging demands (β = .28, p < .001) than optimism was on JC structural resources (β = .14, p < .01) and JC challenging demands (β = .14, p < .05). Furthermore, results showed that empowering leadership (β = .51, p < .001) was the only significant contributor to JC social resources. Optimism (β = −.12, p < .05) was the only significant contributor to JC hindrance demands.

In Hypotheses 3a and 3b, we proposed that optimism moderates the relationship between empowering leadership and JC. More

TABLE 1 Descriptive statistics and Pearson correlations in the study

Measure	α	*M*	*SD*	1	2	3	4	5	6	7
1. JC Structural Resources	.80	4.04	0.60	-						
2. JC Social Resources	.75	2.93	0.69	.40**	-					
3. JC Challenging Demands	.79	3.20	0.77	.51**	.30**	-				
4. JC Hindrance Demands	.80	2.32	0.73	.04	.15**	.10	-			
5. JC Total	.84	3.39	0.53	.79**	.73**	.80**	.13*	-		
6. Empowering Leadership	.86	3.30	0.98	.42**	.52**	.34**	.02	.54**	-	
7. Optimism	.80	3.74	0.72	.23**	.17**	.21**	−.11	.26**	.30**	-

Note. JC = job crafting. N = 330.

*p < .05. **p < .01.

TABLE 2 Goodness of fit indicators of the original four-factor model and three alternative models

Models	χ^2 (*df*)	CFI	TLI	GFI	RMSEA
One-factor	χ^2 (184) = 1351.36***	0.54	0.47	0.68	0.14
Two-factor	χ^2 (183) = 1085.53***	0.77	0.64	0.74	0.13
Three-factor	χ^2 (181) = 787.84***	0.76	0.72	0.80	0.10
Four-factor	χ^2 (178) = 367.76***	0.93	0.91	0.90	0.06

Note. CFI = Bentler comparative fit index; TLI = Tuckler Lewis index; GFI = goodness of fit index; RMSEA = root-mean-square error approximation.

***p < .001.

specifically, we hypothesized that the relationship between empowering leadership and JC would be stronger for employees high (vs. low) in optimism. Table 3 shows the results of the analyses, and Figures 1, 2 depict the resulting plots for the significant interactions. Two interaction effects were in the expected direction as optimism strengthened the relation between empowering leadership and two JC strategies. Figure 1 illustrates the pattern of the interaction effect Empowering Leadership × Optimism (β = .12, p < .05) on JC structural resources. The higher the level of optimism, the stronger was the association between empowering leadership and JC structural resources (Figure 1). A simple slope test was conducted and confirmed that the slopes for high and low optimism were significantly different from each other. We tested different values (0, 1, 5) of the moderator and all became significant, for instance, for moderator value 1, t = 3.01, p = .003. Hypothesis 3a was partly supported. There was no support for an interaction effect on JC increasing social support.

Figure 2 illustrates the interaction effect of Empowering Leadership × Optimism on JC challenging demands (β = .12, p < .05). Again, we see that the higher the level of optimism, the stronger the relation between empowering leadership and JC challenging demands. The simple slope test confirmed that the slopes for high and low optimism were significantly different from each other. We tested different values of the moderator (0, 1, 5) and all became significant, for instance, for moderator value 1, t = 3.20, p = .002. Hypothesis 3b was partly supported. There was no support for an interaction effect on JC decreasing challenging demands.

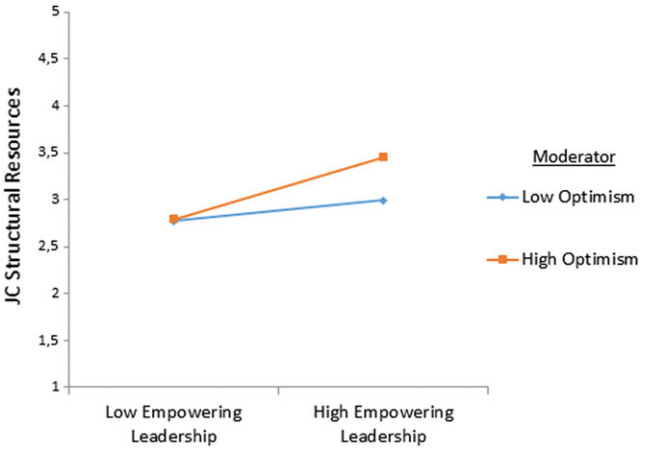


FIGURE 1 Interaction effect of Empowering Leadership × Optimism on job crafting (JC) structural resources

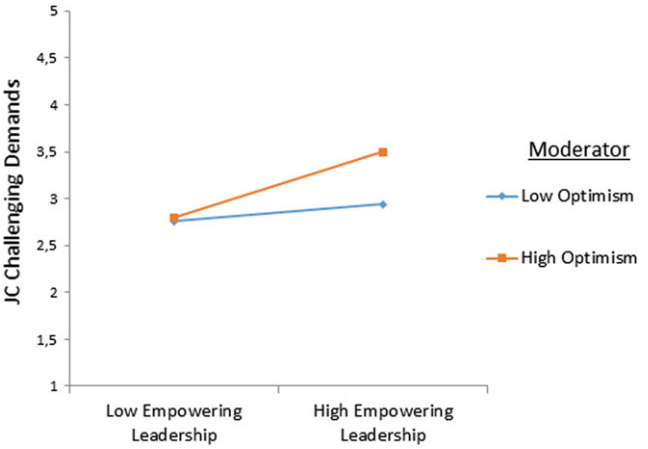


FIGURE 2 Interaction effect of Empowering Leadership × Optimism on job crafting (jc) challenging demands

8 | DISCUSSION

The present study aimed to investigate the relationship between empowering leadership and four JC strategies. Furthermore, we wanted to examine how high versus low optimism would affect the relationship between empowering leadership and the four JC strategies. The main findings of this study contribute to the JC literature by showing that leaders may encourage three JC strategies through empowering leadership and that optimism moderates the influence of empowering leadership on two of these JC strategies.

Leaders' unique ability to inspire and instruct employees to make their jobs more meaningful and engaging (Bakker & Demerouti, 2014; Bakker & Demerouti, 2017) can explain why empowering leadership relates positively to JC. This study offers evidence that empowering

TABLE 3 Structural equation modelling predicting job crafting (JC) from empowering leadership and optimism (n = 308)

	JC structural resources		JC social resources		JC challenging demands		JC hindering demands	
Predictor	b (*SE*)	β	b (*SE*)	β	b (*SE*)	β	b (*SE*)	β
Empowering Leadership	.22 (.03)	.36***	.36 (.04)	.51***	.22 (.05)	.28***	0.3 (.05)	.04
Optimism	.12 (.05)	.14**	.03 (0.5)	.03	.15 (.06)	.14*	−.12 (.06)	−.12*
Empowering Leadership × Optimism	.11 (.05)	.12*	.05 (.05)	.05	.13 (.06)	.12*	.07 (.06)	.05
Total R^2	20.3%		27.9%		14.3%		1.9%	

Note. b = unstandardized path coefficient; β = standardized path coefficient. Fit indices: χ^2 = 3.44, df = 2, χ^2/df = 1.72, TLI = 0.935, CFI = 0.995, RMSEA = 0.047.

*p < .05. **p < .01. ***p < .001.

leadership relates positively to employee JC increasing structural resources, increasing social resources, and increasing challenging demands. The manner in which employees perceive and experience their leaders has implications on how they solve challenges during the working day, indicated by the strong positive association between empowering leadership and three JC strategies. Empowering leadership was the only significant contributor to increasing social resources. This can be explained by leaders' encouragement affecting employees' sense of belonging, which makes the employees confident to take action and to seek out social relations and feedback to optimize their daily work environment (Ahearne et al., 2005; Zhang & Bartol, 2010). For instance, when receiving positive feedback on their JC behaviours, employees may be encouraged to also use JC in the future (Wrzesniewski & Dutton, 2001). The empowering leader may also create an active learning environment (Petrou et al., 2012), and, by giving their employees autonomy and time to develop themselves, leaders show that employees can make their own job more engaging by using JC. Accordingly, empowering leaders inspire employees to optimize their work and social environment, for instance, by asking leaders for feedback, learning new skills, and by asking colleagues for advice (Sims et al., 2009; Zhang & Bartol, 2010).

Our findings suggest that empowering leadership is an important antecedent of three JC strategies. Leaders who empower their followers seem to encourage followers to self-manage their work by using these strategies, especially to search for social resources, structural resources, and to seek challenges (cf. Bakker, 2017). Only one of the predictors in this study had significant contributions to decreasing hindering demands, however, in the opposite direction to what we hypothesized. Optimism had a negative association with decreasing hindering demands. Previous research has shown that decreasing hindering demands is often related to poorer employee well-being (Demerouti, 2014), and it has also been shown to predict exhaustion (Petrou, Demerouti, & Schaufeli, 2015). This strategy seems to have different effects from the other three strategies, and different mechanisms may be involved (Tims et al., 2012). This would be an important matter for further investigation.

Another goal of this study was to examine the moderating role of optimism in the empowering leadership–JC link. Our findings suggest that optimism affects the relationship between empowering leadership and two employee JC strategies, increasing structural resources and increasing challenging demands. The findings indicate that employees' level of optimism affects how employees respond to leaders' empowering leadership behaviours. According to JD–R theory, every work environment has its unique demands and resources (Bakker & Demerouti, 2014, Bakker & Demerouti, 2017), and findings from this study indicate that optimism is a unique personal resource which affects how we use empowering leadership to deal with challenging job demands and mobilize structural job resources in an empowering environment. Taking this into consideration, the moderating role of optimism proposes a tailor-made understanding of JC as it shows that leadership style needs to interact with employees' personal resources to effectively encourage employees to utilize JC in their work environment.

Furthermore, our findings support the notion that personal resources such as optimism can help individuals to deal with challenges and can boost the impact of challenging job demands (Bakker & Sanz-Vergel, 2013). Employees high in optimism proactively seek new work tasks. This can be explained by optimistic individuals having more confidence in their abilities (Bono & Judge, 2003; Carver & Scheier, 2014) and therefore responding with action when a leader encourages JC structural resources and challenging demands. The current findings contribute to the JC literature by illustrating the moderating role of a personal resource and by showing that the relationship between empowering leadership and increasing structural resources and challenging demands is stronger for those high (vs. low) in optimism.

Together with Esteves and Lopes (2017), Slemp et al. (2015), and Wang and colleagues (Wang, Demerouti, & Bakker, 2017), this study provides a leadership perspective that contributes to further development of the JC literature. It might be easier for employees to use the JC strategies of seeking challenges and resources when they have a leader encouraging self-efficacy and self-confidence as empowering leaders aim to do (Ahearne et al., 2005). Although JC concerns self-initiated actions to change job demands and job resources, van Wingerden, Derks, and Bakker (2017) argue that management may support and facilitate the use of JC strategies. By offering empirical evidence that empowering leadership may increase employees' search for structural resources and challenging demands, our findings strengthen the use of JD–R theory as a theoretical framework for understanding how the characteristics of the work environment have implications on the use of JC.

The present study has important practical implications. Specifically, the findings offer information that may enable organizations to promote JC among their employees. As JC is positively related to favourable organizational outcomes such as work engagement and job performance (Bakker et al., 2012; Petrou et al., 2012), information about JC antecedents and gaining knowledge of JC can be encouraged. Also, the link between leadership and JC highlights the importance of focusing on empowering leadership in JC interventions aiming at increasing job resources and challenging demands. Additionally, the optimistic nature of employees may affect the success of such interventions. Interventions should consider personal resources when designing how to promote JC by increasing structural resources and challenging demands.

Although this study provides evidence for most of the hypotheses, some limitations of the study should be mentioned. The cross-sectional design of this study does not allow inferences regarding causality. Future studies should investigate the association between leadership, personal resources, and JC using a longitudinal study with more than two waves to learn more about the leader's role and how to prevent negative consequences of JC and how to promote positive behaviour. Future studies should also investigate whether education level, as well as organizational and job tenure, affects JC. Furthermore, common method variance is a potential problem because of the self-reported nature of all variables. To limit methodological bias, assuring respondents' anonymity decreases strategic responding and increases valid answers (Podsakoff et al., 2003). However, it has been shown that common method bias makes it more difficult to identify interaction effects (Siemsen, Roth, & Oliveira, 2009). The finding of significant interaction effects suggest that common method bias was not a major problem for the current study. Additionally, the adequate

sample size and high response rate of 92% decreases the likelihood of random errors and increases the accuracy of the estimates.

9 | CONCLUSION

Empowering leadership, in other words, empowering employees to manage themselves, is an important antecedent of JC structural resources, social resources, and challenging demands. Optimism and empowering leadership are especially important in JC structural resources and challenging demands. Participants high (vs. low) in optimism are more open to empowering leaders' encouragement and therefore seem to increase their structural job resources and challenging job demands. In order to optimize employees' work environment through JC, organizations may want to invest in empowering leadership.

ACKNOWLEDGEMENTS

The authors are grateful to the assistants responsible for the data collection and recruitment of the participants.

CONFLICT OF INTEREST

We declare no conflict of interest.

DATA ACCESSIBILITY STATEMENT

All data and materials are available upon request to corresponding author.

ORCID

Sylvi Thun http://orcid.org/0000-0001-6275-0112

REFERENCES

Ahearne, M., Mathieu, J., & Rapp, A. (2005). To empower or not to empower your sales force? An empirical examination of the influence of leadership empowerment behavior on customer satisfaction and performance. *Journal of Applied Psychology*, *90*, 945–955. https://doi.org/10.1037/0021-9010.90.5.945

Alarcon, G. M., Bowling, N. A., & Khazon, S. (2013). Great expectations: A meta-analytic examination of optimism and hope. *Personality and Individual Differences*, *54*, 821–827. https://doi.org/10.1016/j.paid.2012.12.004

Bakker, A. B. (2017). Strategic and proactive approaches to work engagement. *Organizational Dynamics*, *46*, 67–75. https://doi.org/10.1016/j.orgdyn.2017.04.002

Bakker, A. B., & Demerouti, E. (2007). The job demands-resources model: State of the art. *Journal of Managerial Psychology*, *22*, 309–328. https://doi.org/10.1108/02683940710733115

Bakker, A. B., & Demerouti, E. (2014). Job demands-resources theory. In P. Y. Chen, & C. L. Cooper (Eds.), *Work and wellbeing: A complete reference guide* (Vol. III (pp. 1–28). John Wiley & Sons.

Bakker, A. B., & Demerouti, E. (2017). Job demands–resources theory: Taking stock and looking forward. *Journal of Occupational Health Psychology*, *22*, 273–285. https://doi.org/10.1037/ocp0000056

Bakker, A. B., & Sanz-Vergel, A. I. (2013). Weekly work engagement and flourishing: The role of hindrance and challenge job demands. *Journal of Vocational Behavior*, *83*, 397–409. https://doi.org/10.1016/j.jvb.2013.06.008

Bakker, A. B., Tims, M., & Derks, D. (2012). Proactive personality and job performance: The role of job crafting and work engagement. *Human Relations*, *65*, 1359–1378. https://doi.org/10.1177/0018726712453471

Bandura, A. (2000). Exercise of human agency through collective efficacy. *Current Directions in Psychological Science*, *9*, 75–78. https://doi.org/10.1111/1467-8721.00064

Bipp, T., & Demerouti, E. (2015). Which employees craft their jobs and how? Basic dimensions of personality and employees' job crafting behaviour. *Journal of Occupational and Organizational Psychology*, *88*, 631–655.

Bono, J. E., & Judge, T. A. (2003). Core self-evaluations: A review of the trait and its role in job satisfaction and job performance. *European Journal of Personality*, *17*, 5–18. https://doi.org/10.1002/per.481

Brislin, R. W. (1970). Back-translation for cross-cultural research. *Journal of Cross-Cultural Psychology*, *1*, 185–216. https://doi.org/10.1177/135910457000100301

Brydon, L., Walker, C., Wawrzyniak, A. J., Chart, H., & Steptoe, A. (2009). Dispositional optimism and stress-induced changes in immunity and negative mood. *Brain, Behavior, and Immunity*, *23*, 810–816. https://doi.org/10.1016/j.bbi.2009.02.018

Byrne, B. M. (2016). *Structural equation modeling with Amos: Basic concepts, applications, and programming*. London: Taylor & Francis.

Carver, C. S., & Scheier, M. F. (2014). Dispositional optimism. *Trends in Cognitive Sciences*, *18*, 293–299. https://doi.org/10.1016/j.tics.2014.02.003

Carver, C. S., Scheier, M. F., & Segerstrom, S. C. (2010). Optimism. *Clinical Psychology Review*, *30*, 879–889. https://doi.org/10.1016/j.cpr.2010.01.006

Clegg, C., & Spencer, C. (2007). A circular and dynamic model of the process of job design. *Journal of Occupational and Organizational Psychology*, *80*, 321–339. https://doi.org/10.1348/096317906X113211

Demerouti, E. (2014). Design your own job through job crafting. *European Psychologist*, *19*, 237–247. https://doi.org/10.1027/1016-9040/a000188

Demerouti, E., Bakker, A. B., Nachreiner, F., & Schaufeli, W. B. (2001). The job demands-resources model of burnout. *Journal of Applied Psychology*, *86*, 499–512. https://doi.org/10.1037//0021-9010.86.3.499

Esteves, T., & Lopes, M. P. (2017). Leading to crafting: The relation between leadership perception and nurses' job crafting. *Western Journal of Nursing Research*, *39*, 763–783. https://doi.org/10.1177/0193945916659507

Gagné, M., & Deci, E. L. (2005). Self-determination theory and work motivation. *Journal of Organizational Behavior*, *26*, 331–362. https://doi.org/10.1002/job.322

Gordon, H. J., Demerouti, E., LeBlanc, P. M., Bakker, A. B., Bipp, T., & Verhagen, M. A. (2018). Individual job redesign: Job crafting interventions in health care. *Journal of Vocational Behavior*, *104*, 98–114.

Grant, A. M., & Ashford, S. J. (2008). The dynamics of proactivity at work. *Research in Organizational Behavior*, *28*, 3–34. https://doi.org/10.1016/j.riob.2008.04.002

Grant, A. M., & Parker, S. K. (2009). Redesigning work design theories: The rise of relational and proactive perspectives. *Academy of Management Annals*, *3*, 317–375. https://doi.org/10.1080/19416520903047327

Hobfoll, S. E. (1989). Conservation of resources: A new attempt at conceptualizing stress. *American Psychologist*, *44*, 513–524. https://doi.org/10.1037/0003-066X.44.3.513

Hobfoll, S. E., Johnson, R. J., Ennis, N., & Jackson, A. P. (2003). Resource loss, resource gain, and emotional outcomes among inner city women. *Journal of Personality and Social Psychology*, *84*, 632–643. https://doi.org/10.1037/0022-3514.84.3.632

Hu, L., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling: A Multidisciplinary Journal*, *6*, 1–55.

Judge, T. A., Locke, E. A., & Durham, C. C. (1997). The dispositional causes of job satisfaction: A core evaluations approach. *Research in Organizational Behavior*, *19*, 151–188.

Krok, D. (2015). The mediating role of optimism in the relations between sense of coherence, subjective and psychological well-being among late adolescents. *Personality and Individual Differences*, *85*, 134–139. https://doi.org/10.1016/j.paid.2015.05.006

Kulik, C. T., Oldham, G. R., & Hackman, J. R. (1987). Work design as an approach to person-environment fit. *Journal of Vocational Behavior*, *31*, 278–296. https://doi.org/10.1016/0001-8791(87)90044-3

Lyons, P. (2008). The crafting of jobs and individual differences. *Journal of Business and Psychology*, *23*, 25–36. https://doi.org/10.1007/s10869-008-9080-2

Martin, S. L., Liao, H., & Campbell, E. M. (2013). Directive versus empowering leadership: A field experiment comparing impacts on task proficiency and proactivity. *Academy of Management Journal*, *56*, 1372–1395. https://doi.org/10.5465/amj.2011.0113

Monzani, D., Steca, P., Greco, A., D'Addario, M., Pancani, L., & Cappelletti, E. (2015). Effective pursuit of personal goals: The fostering effect of dispositional optimism on goal commitment and goal progress. *Personality and Individual Differences*, *82*, 203–214. https://doi.org/10.1016/j.paid.2015.03.019

Pearce, C. L., Sims, H. P., & Kivlighan, D. M. (2002). Vertical versus shared leadership as predictors of the effectiveness of change management teams: An examination of aversive, directive, transactional, transformational, and empowering leader behaviors. *Group Dynamics: Theory, Research, and Practice*, *6*, 172–197. https://doi.org/10.1037/1089-2699.6.2.172

Petrou, P., Demerouti, E., Peeters, M. C. W., Schaufeli, W. B., & Hetland, J. (2012). Crafting a job on a daily basis: Contextual correlates and the link to work engagement. *Journal of Organizational Behavior*, *33*, 1120–1141. https://doi.org/10.1002/job.1783

Petrou, P., Demerouti, E., & Schaufeli, W. B. (2015). Job crafting in changing organizations: Antecedents and implications for exhaustion and performance. *Journal of Occupational Health Psychology*, *20*, 470–480. https://doi.org/10.1037/a0039003

Podsakoff, P. M., MacKenzie, S. B., Lee, J.-Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology*, *88*, 879–903. https://doi.org/10.1037/0021-9010.88.5.879

Rasmussen, H., Scheier, M., & Greenhouse, J. (2009). Optimism and physical health: A meta-analytic review. *Annals of Behavioral Medicine*, *37*, 239–256. https://doi.org/10.1007/s12160-009-9111-x

Rudolph, C. W., Katz, I. M., Lavigne, K. N., & Zacher, H. (2017). Job crafting: A meta-analysis of relationships with individual differences, job characteristics, and work outcomes. *Journal of Vocational Behavior*, *102*, 112–138. https://doi.org/10.1016/j.jvb.2017.05.008

Schaufeli, W. B., & Taris, T. (2014). A critical review of the job demands-resources model: Implications for improving work and health. In G. F. Bauer, & O. Hämmig (Eds.), *Bridging occupational, organizational and public health: A transdisciplinary approach* (1st ed.) (pp. 43–68). Dordrecht, The Netherlands: Springer.

Scheier, M. F., & Carver, C. S. (1985). Optimism, coping, and health: Assessment and implications of generalized outcome expectancies. *Health Psychology*, *4*, 219–247. https://doi.org/10.1037/0278-6133.4.3.219

Scheier, M. F., & Carver, C. S. (1992). Effects of optimism on psychological and physical well-being: Theoretical overview and empirical update. *Cognitive Therapy and Research*, *16*, 201–228. https://doi.org/10.1007/BF01173489

Scheier, M. F., Carver, C. S., & Bridges, M. W. (1994). Distinguishing optimism from neuroticism (and trait anxiety, self-mastery, and self-esteem): A reevaluation of the Life Orientation Test. *Journal of Personality and Social Psychology*, *67*, 1063–1078. https://doi.org/10.1037/0022-3514.67.6.1063

Siemsen, E., Roth, A., & Oliveira, P. (2009). Common method bias in regression models with linear, quadratic, and interaction effects. *Organizational Research Methods*, *13*, 456–476. https://doi.org/10.1177/1094428109351241

Sims, H. P., Faraj, S., & Yun, S. (2009). When should a leader be directive or empowering? How to develop your own situational theory of leadership. *Business Horizons*, *52*, 149–158. https://doi.org/10.1016/j.bushor.2008.10.002

Slemp, G., Kern, M., & Vella-Brodrick, D. (2015). Workplace well-being: The role of job crafting and autonomy support. *Psychology of Well-Being*, *5*, 1–17. https://doi.org/10.1186/s13612-015-0034-y

Spreitzer, G. (1995). Psychological empowerment in the workplace: Dimensions, measurement, and validation. *Academy of Management Journal*, *38*, 1442–1464. https://doi.org/10.2307/256865

Statistics Norway (2015). Labour force survey. Received from http://www.ssb.no/arbeid-og-lonn/artikler-og-publikasjoner/dokumentasjon-av-arbeidskraftundersokelsen-aku-etter-omleggingen-i-2006

Tims, M., & Bakker, A. B. (2010). Job crafting: Towards a new model of individual job redesign. *South African Journal of Industrial Psychology*, *36*, 1–9.

Tims, M., Bakker, A. B., & Derks, D. (2012). Development and validation of the job crafting scale. *Journal of Vocational Behavior*, *80*, 173–186. https://doi.org/10.1016/j.jvb.2011.05.009

Tims, M., Bakker, A. B., & Derks, D. (2014). Daily job crafting and the self-efficacy performance relationship. *Journal of Managerial Psychology*, *29*, 490–507.

Tims, M., Derks, D., & Bakker, A. B. (2016). Job crafting and its relationships with person-job fit and meaningfulness: A three-wave study. *Journal of Vocational Behavior*, *92*, 44–53. https://doi.org/10.1016/j.jvb.2015.11.007

van den Heuvel, M., Demerouti, E., & Peeters, M. C. W. (2015). The job crafting intervention: Effects on job resources, self-efficacy, and affective well-being. *Journal of Occupational and Organizational Psychology*, *88*, 511–532. https://doi.org/10.1111/joop.12128

van Wingerden, J., Bakker, A. B., & Derks, D. (2017). The longitudinal impact of a job crafting intervention. *European Journal of Work and Organizational Psychology*, *26*, 107–119. https://doi.org/10.1080/1359432X.2016.1224233

van Wingerden, J., Derks, D., & Bakker, A. B. (2017). The impact of personal resources and job crafting interventions on work engagement and performance. *Human Resource Management*, *56*, 51–67. https://doi.org/10.1002/hrm.21758

Wang, H.-J., Demerouti, E., & Bakker, A. B. (2017). A review of job crafting research: The role of leader behaviors in cultivating successful job crafters. In S. K. Parker, & U. K. Bindl (Eds.), *Proactivity at work. Making things happen in organizations*. New York, NY: Routledge.

Wang, H.-J., Demerouti, E., & LeBlanc, P. (2017). Transformational leadership, adaptability, and job crafting: The moderating role of organizational identification. *Journal of Vocational Behavior*, *100*, 185–195. https://doi.org/10.1016/j.jvb.2017.03.009

Wannström, I., Peterson, U., Asberg, M., Nygren, A., & Gustavsson, P. (2009). Psychometric properties of scales in the General Nordic Questionnaire for Psychological and Social Factors at Work (QPS): Confirmatory factor analysis and prediction of certified long-term sickness absence. *Scandinavian Journal of Psychology*, *50*, 231–244. https://doi.org/10.1111/j.1467-9450.2008.00697.x

Wrzesniewski, A., & Dutton, J. E. (2001). Crafting a job: Revisioning employees as active crafters of their work. *Academy of Management Review*, *26*, 179–201.

Xanthopoulou, D., Bakker, A. B., Demerouti, E., & Schaufeli, W. B. (2009). Reciprocal relationships between job resources, personal resources, and work engagement. *Journal of Vocational Behavior*, *74*, 235–244. https://doi.org/10.1016/j.jvb.2008.11.003

Zhang, X., & Bartol, K. M. (2010). Linking empowering leadership and employee creativity: The influence of psychological empowerment, intrinsic motivation, and creative process engagement. *Academy of Management Journal*, *53*, 107–128.

How to cite this article: Thun S, Bakker AB. Empowering leadership and job crafting: The role of employee optimism. *Stress and Health*. 2018;34:573–581. https://doi.org/10.1002/smi.2818